



दक्षिण बिहार केन्द्रीय विश्वविद्यालय CENTRAL UNIVERSITY OF SOUTH BIHAR

Question Paper of End-Term Examination, May 2025

Name of Department : Department of Computer Science Programme : M.Sc. (AI)
Session : 2024-2025 Semester : II
Course Title : Research Methodology Course Code : CAI82DC01104
Course Credit : 4 Max. Marks : 70 Duration : 3 hours
Enrolment No. (to be filled by Student) : CUSB2402332028

Instructions: This question paper have three sections. The use of scientific calculator is permitted in this examination.

Section A

This section has ten questions of one point each. All questions are compulsory.

- Q1. What do you mean by term *Ex post facto* research?
- Q2. What are the steps in research process that should be completed before starting the development of the *working hypothesis*?
- Q3. Mention any two tools for collecting the primary data.
- Q4. What do you understand by the term *sampling design*?
- Q5. Write the meaning of *Experimental Group*.
- Q6. Write the name of the statistical tools used in *Measures of Central Tendency*.
- Q7. Define the term *Simulation*.
- Q8. Compute the *Geometric Mean* of the numbers 4, 6 and 9.
- Q9. What is purpose of footnotes in report writing?
- Q10. State the significance of *Report Writing*.

Section B

This section has six questions of six points each. Attempt any five questions.

Q1. . Discuss the following measures of Dispersion

[3+3]

I. Mean Deviation

II. Standard Deviation

Q2. A population is divided into three strata so that $N_1 = 5000$, $N_2 = 2000$ and $N_3 = 3000$ respectively. Respective standard deviations are:

$$\sigma_1 = 15, \sigma_2 = 18 \text{ and } \sigma_3 = 5$$

How should a sample of size $n = 84$ be allocated to the three strata, if we want optimum allocation using disproportionate sampling?

Q3. Explain the following terms in the observation method of data collection.

[2+2+2]

I. Participant Observation & Non-Participant Observation

II. Structured and Unstructured Observation

III. Controlled and Uncontrolled Observation

Q4. The foreman of ABC mining company has estimated the average quantity of iron ore extracted to be 36.8 tons per shift and the sample standard deviation to be 2.8 tons per shift, based upon a random selection of 4 shifts. Construct a 90 percent confidence interval around this estimate by using student's t -distribution. [Critical values tables are attached here for your reference].

Q5. Write a short note on the modelling and simulation and its usability.

Q6. Explain the following IPR

[3+3]

I. Patents

II. Geographical Indication

Section C

This section has three questions of fifteen points each. Attempt any two questions.

Q1. Discuss the difference between one tailed and two tailed hypothesis testing with the help of suitable examples.

Solve the following question.

A marketing company claims that it receives 8% of responses from its mailing. It seems that the company is giving an exaggerated picture and the said percentage is less. To test this claim, a random sample of 500 was surveyed with 30 responses. Test the company's claim at the $\alpha = 0.05$ significance level against the company is giving the exaggerated picture [Critical values tables are attached here for your reference]. [5+10]

Q2. Discuss the following formal research designs with the help of the neat and clean diagram. [3×5]

I. C.R. Design: Two group simple randomized design

II. C.R. Design: Random Replication Design

III. L.S. Design: Latin Square Design

Q3. Discuss the following in the Layout of research report [4+7+4]

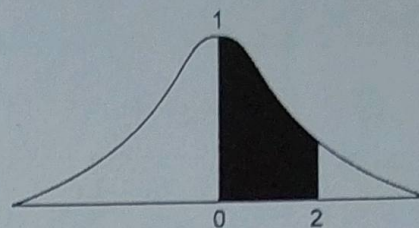
I. Preliminary Pages

II. Main Text

III. End Matter

Table 1: Area Under Normal Curve

An entry in the table is the proportion under the entire curve which is between $z = 0$ and a positive value of z . Areas for negative values for z are obtained by symmetry.



Areas of a standard normal distribution

z	.0	0.01	.02	.03	.04	.05	.06	.07	.08	.09
.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
.7	.2580	.2611	.2642	.2673	.2703	.2734	.2764	.2794	.2823	.2852
.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Table 2: Critical Values of Student's *t*-Distribution

d.f.	Level of significance for two-tailed test					d.f.
	0.20	0.10	0.05	0.02	0.01	
	Level of significance for one-tailed test					
	0.10	0.05	0.025	0.01	0.005	
1	3.078	6.314	12.706	31.821	63.657	1
2	1.886	2.920	4.303	6.965	9.925	2
3	1.638	2.353	3.182	4.541	5.841	3
4	1.533	2.132	2.776	3.747	4.604	4
5	1.476	2.015	2.571	3.365	4.032	5
6	1.440	1.943	2.447	3.143	3.707	6
7	1.415	1.895	2.365	2.998	3.499	7
8	1.397	1.860	2.306	2.896	3.355	8
9	1.383	1.833	2.262	2.821	3.250	9
10	1.372	1.812	2.228	2.764	3.169	10
11	1.363	1.796	2.201	2.718	3.106	11
12	1.356	1.782	2.179	2.681	3.055	12
13	1.350	1.771	2.160	2.650	3.012	13
14	1.345	1.761	2.145	2.624	2.977	14
15	1.341	1.753	2.131	2.602	2.947	15
16	1.337	1.746	2.120	2.583	2.921	16
17	1.333	1.740	2.110	2.567	2.898	17
18	1.330	1.734	2.101	2.552	2.878	18
19	1.328	1.729	2.093	2.539	2.861	19
20	1.325	1.725	2.086	2.528	2.845	20
21	1.323	1.721	2.080	2.518	2.831	21
22	1.321	1.717	2.074	2.508	2.819	22
23	1.319	1.714	2.069	2.500	2.807	23
24	1.318	1.711	2.064	2.492	2.797	24
25	1.316	1.708	2.060	2.485	2.787	25
26	1.315	1.706	2.056	2.479	2.779	26
27	1.314	1.703	2.052	2.473	2.771	27
28	1.313	1.701	2.048	2.467	2.763	28
29	1.311	1.699	2.045	2.462	2.756	29
Infinity	1.282	1.645	1.960	2.326	2.576	Infinity